

The fourth row

No ultraviolet divergence — and the p^+ integral in closed form

$$A \equiv (x - z)^2, \quad B \equiv (x - w)^2, \quad D \equiv (\vee - k^+)A + k^+B$$

0. Verdict

The expression is correct — prefactor, phases and the way the residual $\frac{1}{p^+ + k^+}$ is absorbed into $\mathfrak{B}_2$'s bracket all check out exactly (§1).

No transverse ultraviolet divergence. Every coincidence is regular — in fact more regular than merely integrable (§2). The reason is that *there is no $\mathfrak{C}$ in this row*: last time the UV came from $\mathfrak{C}$'s kernel meeting $\mathfrak{B}_2$'s squared denominator on the *same* separation, and here that pairing does not exist.

There is an infrared divergence at large $|z|$ (§3) — z is the one variable carrying no phase.

The p^+ integral is elementary (no $\mathfrak{C}$ means no δ -function and hence no p^+ -dependent phase) and comes out as Weizsäcker–Williams kernels times logarithms (§4).

1. The expression

item	from the definitions	quoted
phases	$(-i)(-i)(+i) = -i$	$-i \checkmark$
numeric	$\frac{2}{(2\pi)^3} \cdot \frac{1}{(\sqrt{2\pi})^2} \cdot \frac{1}{4\pi^2} = \frac{1}{32\pi^7}$	$\frac{1}{(2\pi)^3} \frac{1}{4\pi^4} = \frac{1}{32\pi^7} \checkmark$
k^+	$\frac{1}{\sqrt{k^+}} \frac{1}{\sqrt{p^+}} \cdot \frac{\sqrt{p^+ k^+}}{p^+ + k^+} = \frac{1}{p^+ + k^+}$	absorbed — see below $\checkmark$

That leftover $\frac{1}{p^+ + k^+}$ divides $\mathfrak{B}_2$'s bracket, turning

$$\left[\frac{\delta^{ij}(A-B)}{2(z-w)^2} + \frac{p^+ + k^+}{k^+}(\dots) + \frac{p^+ + k^+}{p^+}(\dots) \right] \longrightarrow \left[\frac{\delta^{ij}(A-B)}{2(p^+ + k^+)(z-w)^2} + \frac{1}{k^+}(\dots) + \frac{1}{p^+}(\dots) \right],$$

which is exactly the bracket you wrote, with no overall k^+ power left over. Difference from the quoted prefactor: identically zero.

2. No ultraviolet divergence

Measured $\langle \text{integrand} \rangle_\theta$ at each coincidence. A finite constant means the integrand is $O(1)$ there — better than the $O(1/\rho)$ that would already suffice:

region	$\langle F \rangle$ at $\rho = 10^{-2} \dots 10^{-5}$	verdict
$x \rightarrow z$	-0.12489 (constant)	$O(1)$ — regular
$x \rightarrow w$	+0.10032 (constant)	$O(1)$ — regular
$z \rightarrow w$	+0.22569 (constant)	$O(1)$ — regular
$y \rightarrow z$	$\rightarrow 0$	regular
$x' \rightarrow w'$	single WW kernel	$O(1/\rho)$ — integrable

The triple collapse $x \sim z \sim w$, where $\mathfrak{B}_2$'s denominator $p^+A + k^+B$ vanishes, does give $\langle F \rangle \sim 1/\rho^2$ ($\rho^2 \langle F \rangle = 0.088647$, stable over four decades). But three collapsing points mean two relative two-vectors, measure $\rho^4 \frac{d\rho}{\rho}$, so the net is $\rho^2 \frac{d\rho}{\rho}$: **convergent**.

Why this row differs from the last one. There the ultraviolet divergence came from $\mathfrak{C}$'s kernel $\frac{(z-x)^m}{(z-x)^2}$ contracting with $\mathfrak{B}_2$'s $\frac{1}{(x-z)^2}$ — the *same* separation twice, hence $1/\rho^2$. This row has no $\mathfrak{C}$ at all: the two $\mathfrak{A}$'s supply kernels on $(x' - w')$ and $(y - z)$, which are different separations from $\mathfrak{B}_2$'s $(z - w)$, $(x - z)$, $(x - w)$. Nothing pairs up.

3. But there is an infrared divergence, at large $|z|$

$\langle F \rangle$ falls as $1/R^2$, and z is the one integration variable carrying *no* phase:

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R=1e+02: R^2<F> = -1.06404661
R=1e+03: R^2<F> = -1.06418776
R=1e+04: R^2<F> = -1.06418917
R=1e+05: R^2<F> = -1.06418919
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so $\int \rho d\rho (c/\rho^2) = c \log R$: logarithmically divergent. Large $|w|$ has the same $1/R^2$ (-0.11792) but w carries $e^{+ik \cdot w}$, whose oscillation regulates it.

4. The p^+ integral

With no $\mathfrak{C}$ there is no δ -function and hence no p^+ -dependent phase: the three integrals are rational and elementary.

$$\begin{aligned}\mathcal{I}_1 &= \int_{\Lambda}^{\vee-k^+} \frac{dp^+}{(p^+ + k^+)(p^+ A + k^+ B)} = \frac{1}{k^+(B-A)} \log \frac{\vee B}{D} \\ \mathcal{I}_2 &= \int_{\Lambda}^{\vee-k^+} \frac{dp^+}{p^+(p^+ A + k^+ B)} = \frac{1}{k^+ B} \left[\log \frac{\vee - k^+}{\Lambda} + \log \frac{k^+ B}{D} \right] \\ \mathcal{I}_3 &= \int_{\Lambda}^{\vee-k^+} \frac{dp^+}{p^+ A + k^+ B} = \frac{1}{A} \log \frac{D}{k^+ B}\end{aligned}$$

(all at $\Lambda \rightarrow 0$; the exact finite- Λ forms are in the results file).

Three things fall out. In $\mathcal{I}_1$ the coefficient $\frac{A-B}{2(z-w)^2}$ cancels the $\frac{1}{B-A}$, so the apparent $A = B$ singularity **disappears**. In $\mathcal{I}_2$ the $\frac{1}{B}$ combines with the $(x-w)^i$ in the numerator into a WW kernel $\frac{(x-w)^i}{(x-w)^2}$, and in $\mathcal{I}_3$ the $\frac{1}{A}$ does the same with $(x-z)^j$.

5. Final answer

$$\begin{aligned}\frac{d^3 N_{\text{NLO}}}{d^3 k} &= -\frac{1}{(2\pi)^3} \frac{ig^4}{8\pi^5} \frac{1}{k^+} \int_{z,w,w',x,y,x'} e^{-ik \cdot (w'-w)} \frac{(x'-w')^i}{(x'-w')^2} \frac{(y-z)^j}{(y-z)^2} \mathcal{T}^{ij} \times [\text{colour}] \\ \mathcal{T}^{ij} &= -\frac{\delta^{ij}}{2(z-w)^2} \log \frac{\vee(x-w)^2}{D} + \frac{(x-w)^i}{(x-w)^2} \left[\frac{(z-w)^j}{(z-w)^2} + \frac{(x-z)^j}{2(x-z)^2} \right] \left[\log \frac{\vee - k^+}{\Lambda} + \log \frac{k^+(x-w)^2}{D} \right] \\ &\quad + \frac{(x-z)^j}{(x-z)^2} \left[\frac{(z-w)^i}{(z-w)^2} - \frac{(x-w)^i}{2(x-w)^2} \right] \log \frac{D}{k^+(x-w)^2}, \quad D = (\vee - k^+)(x-z)^2 + k^+(x-w)^2,\end{aligned}$$

with the colour bracket $[U^{ab'}(x')U^{c'd'}(y)U^{e'b}(z)\rho^{b'}(x')\rho^{d'}(y) - U^{bd'}(z)U^{d'e'}(y)U^{ac'}(w')\rho^{c'}(x')\rho^{e'}(y)] [f^{cbd}U^{ac}(w)\rho^d(x) - f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)]$ unchanged. The prefactor picked up the $\frac{1}{2\pi}$ from $\frac{dp^+}{2\pi}$: $\frac{1}{4\pi^4} \cdot \frac{1}{2\pi} = \frac{1}{8\pi^5}$.

Checked against direct p^+ quadrature on four random geometries and three index pairs, to 10^{-7} – 10^{-9} relative (quadrature-limited).

The two longitudinal endpoint logarithms

- $\log \frac{\vee - k^+}{\Lambda}$, from $\mathcal{I}_2$ — the rapidity log, from $p^+ \rightarrow 0$. It rides on $\frac{(x-w)^i}{(x-w)^2} \left[\frac{(z-w)^j}{(z-w)^2} + \frac{(x-z)^j}{2(x-z)^2} \right]$, a product of WW kernels.
- A $\log \vee$ from $\mathcal{I}_3$ — as $\vee \rightarrow \infty$, $\log \frac{D}{k^+(x-w)^2} \rightarrow \log \frac{\vee(x-z)^2}{k^+(x-w)^2}$. This is a genuine large- p^+ logarithm: the $\frac{1}{k^+}$ bracket term has no p^+ in its numerator, so against $\frac{1}{p^+ A + k^+ B}$ it gives $\int^{\vee} \frac{dp^+}{p^+}$. Worth checking against the kinematic origin of $\vee$ — it is not a rapidity divergence and will not be absorbed by evolution. In $\mathcal{I}_2$ by contrast the $\vee$'s cancel: $\log \frac{\vee}{\Lambda} + \log \frac{k^+ B}{\vee A} = \log \frac{k^+ B}{\Lambda A}$.

zint/src/row4.py, row4b.py, row4_final.py — every number above, written to zint/RESULTS_row4.txt.